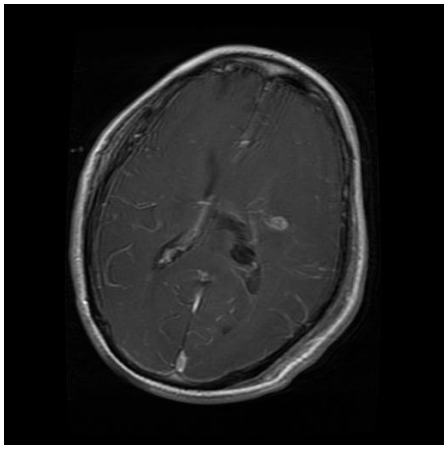
NEURO-AI DIAGNOSTIC CLINICAL REPORT

METAHEURISTIC OPTIMIZED AI
Confidential Medical Document

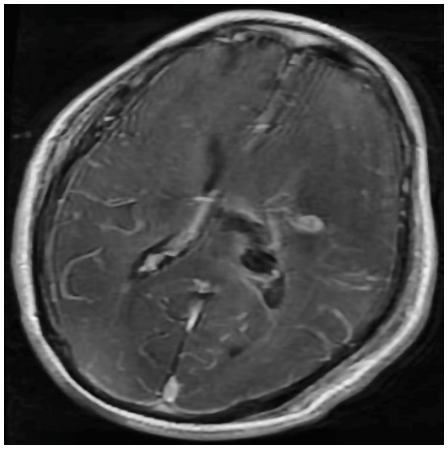
Patient ID: PAT-TEST-001	**Patient Name:** Clinical Test Subject	**Scan Date:** 2026-10-27 09:35:03
Modality: Brain MRI (T1-CE / Axial)	**AI Engine:** Metaheuristic Multi-CNN	**Architecture:** ENSEMBLE

PATHOLOGICAL FINDING: GLIOMA DETECTED (Confidence: 99.3%)

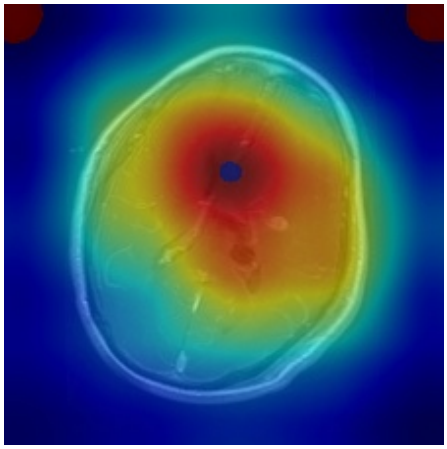
MULTI-STAGE MRI VISUALIZATION & EXPLAINABILITY (XAI)



1. Raw MRI Acquisition
Unprocessed input scan



2. Skull-Stripped & CLAHE
Contour cropped & denoised



3. Grad-CAM++ Saliency
AI Attention Tumor Map

DIAGNOSTIC PROBABILITY DISTRIBUTION & TUMOR LOCALIZATION

Pathology / Class	Probability	Clinical Risk
Glioma	**99.3%**	High
Meningioma	**0.3%**	Moderate
Pituitary Tumor	**0.2%**	Moderate
Healthy / No Tumor	**0.3%**	None

Quantitative Lesion Metric	Observation Value
Estimated Tumor Area Burden	**27.51%** of brain tissue
Anatomical Localization	**Sellar / Central Midline**
Mass Effect Index	**High Mass Effect**
Pathological Description	A primary brain tumor originating in the supportive glial cells of the brain and spinal cord. Typically exhibits irregular, infiltrating margins with surrounding edema.

CLINICAL PROGNOSIS, CURABILITY & PHARMACOTHERAPY RECOMMENDATIONS

ESTIMATED CURABILITY RATE: 15.0% | Chronic / Recurrence Risk: 85.0% | **Prognosis:** Guarded to Moderate (Multimodal Management Required)

Recommended Primary Solution: Maximal Safe Surgical Resection + Adjuvant Chemoradiotherapy (Stupp Protocol)
Surgical / Clinical Pathway: Image-guided craniotomy with intraoperative MRI / 5-ALA fluorescence-guided cortical mapping for

gross-total resection (GTR) of enhancing mass.

Recommended Medicine	Drug Category	Clinical Dosage & Purpose
Temozolomide (TMZ)	Alkylating Oral Chemotherapy (First-Line)	**Dosage:** 75 mg/m²/day orally during 6-week radiotherapy; then maintenance 150–200 mg/m²/day on days 1–5 of 28-day cycles (6–12 cycles) Crosses blood-brain barrier to methylate neoplastic DNA at O6 and N7 positions of guanine, triggering tumor cell apoptosis.
Dexamethasone	High-Potency Corticosteroid	**Dosage:** 4 mg to 8 mg orally / IV every 8 to 12 hours (taper slowly post-resection) Rapidly stabilizes endothelial junctions to reduce life-threatening peritumoral vasogenic edema and intracranial pressure (ICP).
Bevacizumab (Avastin)	Anti-VEGF Monoclonal Antibody	**Dosage:** 10 mg/kg IV infusion every 2 weeks Inhibits vascular endothelial growth factor (VEGF-A) to starve neoplastic microvascular proliferation and relieve refractory edema.
Levetiracetam (Keppra)	Broad-Spectrum Antiepileptic Prophylaxis	**Dosage:** 500 mg to 1000 mg orally twice daily Non-enzyme-inducing anti-convulsant for perioperative seizure prophylaxis without interfering with chemotherapeutic metabolism.

Attending Radiologist / Clinician Sign-Off:

_____

Signature & Stamp

AI Assistance Disclaimer:

This report is generated by a deep learning metaheuristic framework as a clinical decision support tool. Final clinical diagnosis must be verified by a board-certified radiologist.